RAJALAKSHMI ENGINEERING COLLEGE

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R2

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|---|---|
| **Started on** | Friday, 14 August 2026, 9:04 AM |
| **State** | Finished |
| **Completed on** | Friday, 14 August 2026, 9:04 AM |
| **Time taken** | 44 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

// Divide and Conquer function to find the index of the first '0'
int findFirstZero(int arr[], int low, int high, int m) {
    if (high >= low) {
        int mid = low + (high - low) / 2;

        // Check if mid is the first occurrence of 0
        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
            return mid;
        }

        // If element is 1, the first zero must be in the right half
        if (arr[mid] == 1) {
            return findFirstZero(arr, mid + 1, high, m);
        }
        // If element is 0, the first zero could be further left
        else {
            return findFirstZero(arr, low, mid - 1, m);
        }
    }
    return -1; // Returns -1 if no zeroes are present
}

int main() {
    int m;
    if (scanf("%d", &m) != 1) return 0;

    int arr[m];
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    int firstZeroIndex = findFirstZero(arr, 0, m - 1, m);
    int zeroCount = 0;

    if (firstZeroIndex != -1) {
        zeroCount = m - firstZeroIndex;
    } else if (m > 0 && arr[0] == 0) {
        // If all elements are zeroes
        zeroCount = m;
    } else {
        // If all elements are ones
        zeroCount = 0;
    }

    printf("%d\n", zeroCount);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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